QuizSmart - Program Architecture

ARM7 LPC21xx Embedded Interactive Quiz System

System Architecture

QuizSmart follows a modular embedded-system architecture in which the LPC21xx ARM7 microcontroller acts as the central controller. UART0 provides the serial user interface, GPIO handles push-button inputs and LED outputs, and the LCD provides local user instructions.

Module	Interface	Responsibility
UART Terminal	UART0	Question menu, question/options display, question-number input, result display
Four Push Buttons	GPIO inputs	User answer selection: SW1=A, SW2=B, SW3=C, SW4=D
16x2 LCD	LCD Driver	Displays selection instruction to the user
Correct/Wrong LEDs	GPIO outputs	Hardware result indication interface
LPC21xx ARM7	Main Application	Peripheral initialization, quiz control, option detection and answer verification

Block Architecture

UART TERMINAL
Question input and serial output

↓ UART0

LPC21xx ARM7 MICROCONTROLLER
Main quiz application and control logic

■ GPIO / LCD ■

GPIO SWITCHES SW1 P0.15 | SW2 P0.2 | SW3 P0.3 | SW4 P0.4	**16x2 LCD** User instruction display

GPIO output

CORRECT LED P0.6 **WRONG LED** P0.5

Pin Configuration

Signal	LPC21xx Pin	Purpose
SW1	P0.15	Option 1 / A
SW2	P0.2	Option 2 / B
SW3	P0.3	Option 3 / C
SW4	P0.4	Option 4 / D
Wrong LED	P0.5	Incorrect answer indication
Correct LED	P0.6	Correct answer indication

Modular Software Structure

Application Layer: main.c contains the quiz menu, question handling, answer verification and user interaction logic.

Driver Layer: UART, LCD, GPIO and delay routines can be maintained as separate reusable modules.

Hardware Layer: LPC21xx ARM7, UART interface, LCD, push buttons and LEDs.

Documentation: Individual driver and delay documentation can be maintained separately in the repository documentation directory.